On Merton's Portfolio Problem

A Stochastic Optimal Control Problem

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Abstract. The purpose of this thesis is to examine and solve a classic financial optimization problem known as Merton's Portfolio Problem. The problem is driven by a stochastic process and can thereby be classified as a stochastic optimal control problem. The thesis provides the basic theory of how the dynamic programming principle allows dividing the control problem into subproblems that can be linked to a PDE. The solution to the PDE can then be shown to rely on the optimal control sought after in the stochastic control problem. Merton's Portfolio Problem is presented and solved. The explicit solution and the financial implications are then examined, both analytically and by looking at some simulated trajectories.

Sammanfattning. Syftet med den här uppsatsen är att undersöka och lösa ett klassiskt finansiellt optimeringsproblem kallat Mertons Portföljproblem. Problemet drivs av en stokastisk process och kan därför klassas so ett stokastiskt kontrollproblem. Uppsatsen lägger fram grundläggande teori om hur dynamisk programmering kan tillämpas för att dela upp kontrollproblemet i mindre problem som kan knytas till en PDE. Lösningen till PDEn visas bero av den optimala kontrollen som eftersöks i kontrollproblemet. Mertons Portföljproblem presenteras och löses. Den explicita lösningen och dess finansiella implikation undersöks, både analytiskt och genom att titta på några simulerade banor.

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1. Introduction

We live in a world where decision-making is important and present in most of space and time. These decisions could vary in importance, from trivial decisions about the size of a coffee cup to deciding which patient should be on top of the donor list. Decisions could be very complex, but consistently, we seek to optimize the outcome. The optimal outcome might or might not be easily quantifiable, but it is what we strive for when making decisions. One of the most iconic mathematicians in history, Leonhard Euler, supposedly said: "Nothing takes place in the world whose meaning is not that of some maximum or minimum.".

With Euler's quote in mind, this thesis will study the topic of optimizing a stochastic value function based on a stochastic differential equation. The reason to introduce stochasticity to the optimization problem is to handle cases that are either dependent on a truly random event or are so complex and unpredictable that the best approximation of the dynamics is a stochastic model.

We see stochastic optimization problems in many fields. An example from engineering is the optimization of subsystems in a hybrid electric vehicle, see [5]. More specifically to control the drive train in a hybrid electric vehicle such that we maximize efficiency and minimize emissions. While driving at constant speed on a flat and straight road the power demand is close to deterministic, but that is seldom the case in a real scenario. A more realistic driving cycle consists of a lot of variable factors which makes the power demand impossible to predict and stochastic in nature. Controllable processes, in this case, include but are not limited to; choosing gear, choosing the relative distribution of power between the electric engine and the combustion engine, and choosing how much to recharge the battery at any given time. Note that these processes all are depending on the current power demand.

In this thesis, we will look closer at a stochastic optimization problem from the financial sector. The problem is known as Merton's Portfolio Problem, named after Robert C. Merton who first published a solution to the problem in 1969 [6]. The problem can be interpreted and phrased as follows: A working investor who will retire at a known time in the future wants to maximize their pension portfolio, which consists of a combination of a risky and a risk-free asset. The asset including risk is assumed to behave stochastically and the risk-free asset behaves deterministically. The investor also wants to maximize consumption in the time leading up to retirement. The problem consists of two decisions that have to be made, we will consider them as two questions. What is the optimal consumption rate at any given time? What is the optimal balance between the risky and the risk-free asset in the pension portfolio? Note that both these questions affect each other, a higher consumption leaves less in the portfolio and a change in the portfolio changes the future potential consumption.

The purpose of this thesis is to look at and solve the stochastic optimal control problem known as Merton's Portfolio Problem using techniques based on dynamic programming and the Hamilton Jacobi Bellman equation. To do this we will first need to introduce the theory of stochastic calculus and stochastic optimal control

We will start in section 2 by introducing stochastic calculus with key concepts such as the Brownian motion, stochastic integrals, stochastic differential equations, and a powerful result known as Itô's Lemma which we will see be very useful in the following sections.

Section 3 covers some theory of stochastic optimal control. We will see how it is possible, through the Dynamic Programming Principle, to tie the solution of the optimization problem to a deterministic partial differential equation known as the Hamilton-Jacobi-Bellman equation. The Verification Theorem will then show that any solution to the Hamilton-Jacobi-Bellman equation is a solution to the optimal control problem.

After reviewing the technique of solving problems of this sort we will in section 4 specify Merton's portfolio problem and apply the theory from section 3 to find an explicit solution, both in finite and infinite time.

We end this thesis with section 5 by examining the financial implications of the solution and how it relates to the different parameters. This will be done by analyzing partial derivatives of the solution and by looking at simulated paths of the solution.

1.1. **Delimitation.** The theory of stochastic control presented in this thesis is focused on the one-dimensional case. While the technique required for the multidimensional case is similar to the one-dimensional case, the notation gets slightly more cluttered in the multidimensional case as it basically is doing the same calculations multiple times. This can of course be solved with the more compact matrix notation.

All stochastic processes in this text are based on the Brownian Motion, defined in section 2.1. We note that there is possible to use stochastic calculus on stochastic processes that behave differently, a jump process for example has discrete movements. This will however not be covered in this thesis.

We shall also note that there exist other methods to solve similar problems, such as the martingale approach using the Lagrange relaxation, see Chapter 27 in [2]. This thesis will focus on the method of using dynamic programming to connect the problem with a Hamilton-Jacobi-Bellman equation which is a more general method.

2. Stochastic Calculus

In order to solve or even understand the stochastic problems presented in this text, we need to introduce concepts and definitions of stochastic calculus. Stochastic calculus is a part of mathematics under which stochastic processes can be analyzed. The stochastic element of all our calculations will be the Brownian motion, see Definition 2.1. We will see how to integrate over the increments of a Brownian motion and how to interpret the meaning of such an integral. Stochastic differential equations will be introduced as a tool to describe some stochastic processes. We look closer at a result called Itô's Lemma, which could be described as a stochastic version of the chain rule. To finish the preliminaries we look at a connection between stochastic processes and deterministic partial differential equations called the Feynman Kac Formula.

Throughout the text, we consider a fixed probability space $(\Omega, \mathcal{F}, \mathbb{P})$ on which we define a standard Brownian motion $\{W_s\}_s$.

2.1. Brownian Motion.

Definition 2.1 (Definition 4.1 in [2])**.** *The Brownian Motion, W, is a stochastic process defined by*

- $W_0 = 0$.
- *The process W has independent increments, i.e. if $r < s < u$ then $W_u - W_t$ and $W_s - W_r$ are independent random variables.*
- *For $s < t$ the random variable $W_t - W_s$ has the Gaussian distribution $N[0, t-s]$.*
- *W has continuous trajectories.*

We note that $E[W_t] = 0$ and $Var[W_t] = t$, this follows directly from the definition.

A Brownian motion trajectory is everywhere continuous but nowhere differentiable. The fact that it has infinite variation, i.e. it changes infinitely many times at any given time interval, makes it impossible to handle in normal calculus. This is why we introduce stochastic calculus. A Brownian motion will be denoted by $W_t(\omega)$ or W_t, in this thesis, where t is the time of the process and $\omega \in \Omega$ is a single outcome, the randomly chosen path, from the probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Loosely speaking, a **filtration** $\mathcal{F}_t^W$ denotes what has happened to the process W up to time t and we say that a process X_t is $\mathcal{F}_t^W$**-measurable** if we at time t can determine its value given the information generated by W up to time t. If X_t is measurable for all $t \geq 0$ we say that X is **adapted** to the $\mathcal{F}_t^W$-filtration.

2.2. Stochastic Integrals.

A specific stochastic process which we will use throughout this text is the Itô-diffusion, X_t, defined as the solution to a stochastic differential equation, we will look closer at these in section 2.3, and described by its dynamics

$$X_t = X_0 + \int_0^t \mu(s, X_s)ds + \int_0^t \sigma(s, X_s)dW_s$$

where W is a Brownian motion. For simplicity we will use the acknowledged short hand notation

$$dX_t = \mu(t, X_t)\, dt + \sigma(t, X_t)dW_t.$$

We see that the integral above is different from what we are used to in deterministic calculus. The first term, $\int_0^t \mu(s, X_s)ds$, is no problem to integrate using the

Riemann-Stieltjes integral. The second term, $\int_0^t \sigma(s, X_s) dW_s$, however, does not behave as we are used to due to the infinite variations. Another method of integration is needed. We start by defining the class of all functions for which our stochastic integral will be defined.

Definition 2.2 (Definition 3.1.4 in [10])**.** *Assume that g is continuous. Let $\pounds^2 = \pounds^2(S,T)$ be the class of functions*

$$g(t,\omega) : [0,\infty) \times \Omega \to \mathbb{R}]$$

such that

- *$g(t,\omega)$ is $\mathcal{F}_t^W$-adapted.*
- $E\left[\int_S^T g(t,\omega)^2\, dt\right] < \infty$

Since $g \in \pounds^2$, g is adapted to the $\mathcal{F}_t^W$-filtration and the value $g(t_k)$ is only dependent on what has happened up until time t_k. By definition the Brownian motion has independent increments, i.e. the increment $[W_{t_{k+1}} - W_{t_k}]$ is independent of $g(t_k)$. For this reason we will choose to evaluate g at the left end of the interval $[t_k, t_{k+1}]$. This method of stochastic integration is known as Itô-integration. Evaluating at other points on $[t_k, t_{k+1}]$ is mathematically possible but since this would mean looking into the future (the value of $g_{t_{k+1}}$ is unknown at time t_k) the interpretation gets complicated for most applications.

To define the Itô-integral, we start by considering a case with an elementary function $\phi(t,\omega) \in \pounds^2$, defined by

$$\phi(t,\omega) = \sum_{j\geq 0} \phi(t_j,\omega) \cdot \chi_{[t_j,t_{j+1})}(t),$$

where χ is the indicator function and n is a natural number. For an elementary function like this we will define the Itô-integral by

$$\int_S^T \phi(t,\omega)\, dW_t = \sum_{j\geq 0} \phi(t_j,\omega)[W_{t_{j+1}} - W_{t_j}].$$

We will now construct the Itô-integral of $g(t,\omega) \in \pounds^2$ by showing that every $g \in \pounds^2$ can be approximated by elementary functions such as $\phi(t,\omega)$ and then define the Itô-integral of g as the limit of the Itô-integral of ϕ when $\phi \to g$. We will show that the approximation is possible by considering the following three propositions.

Proposition 2.3 (Step 1. in [10] p.27)**.** *For every bounded function $a(t,\omega) \in \pounds^2$ that is continuous for each ω there exists elementary functions $\phi_n(t,\omega)$ such that*

$$E\left[\int_S^T (a - \phi_n)^2\, dt\right] \to 0 \quad \textit{as } n \to \infty.$$

Proof. Define $\phi_n(t,\omega) = \sum_j a(t_j,\omega) \cdot \chi_{[t_j,t_{j+1})}(t)$. Then ϕ_n is elementary since $a \in \pounds^2$, and

$$E\left[\int_S^T (a - \phi_n)^2\, dt\right] \to 0 \quad \text{as } n \to \infty$$

□

Proposition 2.4 (Step 2. in [10] p.27)**.** *If $b(t,\omega) \in \pounds^2$ is bounded, then exist bounded functions $a_n(t,\omega) \in \pounds^2$ that is continuous for all ω and n, such that*

$$E\left[\int_S^T (b - a_n)^2 \, dt\right] \to 0 \quad \text{as } n \to \infty.$$

Proof. Suppose $|b(t,\omega)| \leq M$ for all (t,ω). For each n let ψ_n be a non-negative, continuous function on $\mathbb{R}$ such that

- $\psi_n(x) = 0$ for $x \leq -\frac{1}{n}$ and $x \geq 0$
- $\int_{-\infty}^{\infty} \psi_n(x) \, dx = 1$

Define

$$a_n(t,\omega) = \int_0^t \psi_n(s-t) b(s,\omega) \, ds.$$

Then $|a_n(t,\omega)| \leq M$ and a_n is continuous for each ω. From the fact that $b \in \pounds^2$ it can be shown that a_n is adapted to the $\mathcal{F}_t^W$-filtration. Furthermore,

$$\int_S^T (a_n(s,\omega) - b(s,\omega))^2 \, dt \to 0 \quad \text{as } n \to \infty, \text{ for each } \omega,$$

since $\{\psi_n\}_n$ is an approximate identity. Bounded convergence gives

$$E\left[\int_S^T (b(t,\omega) - a_n(t,\omega))^2\right] \to 0 \quad \text{as } n \to \infty.$$

□

Proposition 2.5 (Step 3. in [10] p.28)**.** *For any function $g(t,\omega) \in \pounds^2$ exists a sequence $\{b_n\} \subset \pounds^2$ such that b_n is bounded for each n and*

$$E\left[\int_S^T (g(t,\omega) - b_n(t,\omega))^2 \, dt\right] \to 0 \quad \text{as } n \to \infty$$

Proof. Put

$$h_n(t,\omega) = \begin{cases} -n & \text{if} \quad f(t,\omega) < -n \\ f(t,\omega) & \text{if} \quad -n \leq f(t,\omega \leq n \\ n & \text{if} \quad f(t,\omega) > n \end{cases}$$

Then follows by the dominated convergence theorem, see [13] Theorem 11.32,

$$E\left[\int_S^T (g(t,\omega) - b_n(t,\omega))^2 \, dt\right] \to 0 \quad \text{as } n \to \infty$$

□

Putting Propositions 2.3, 2.4, and 2.5 together, we see that any function $g \in \pounds^2$ can be approximated by an elementary function and we are now ready to define the Itô-integral for g.

Definition 2.6 (Itô-Integral, Definition 3.1.6 in [10])**.** *Let $g(t,\omega) \in \pounds^2(S,T)$. Then the Itô integral of g from S to T is defined by*

$$\int_S^T g(t,\omega) \, dW_t(\omega) = \lim_{n\to\infty} \int_S^T \phi_n(t,\omega) \, dW_t$$

where $\{\phi_n\}$ is a sequence of elementary functions such that

$$E\left[\int_S^T (g(t,\omega) - \phi_n(t,\omega))^2 \, dt\right] \to 0 \quad \text{as } n \to \infty.$$

The existence of a stochastic integral is fundamental in the sections below containing stochastic differential equations. The Itô-integral comes with a few useful properties.

Proposition 2.7 (Proposition 4.5 in [2])**.** *Let g be a stochastic process in $\pounds^2$, then the following properties hold:*
(i) $E\left[\int_S^T g_t \, dW_t\right] = 0$
(ii) $E\left[\left(\int_S^T g_t \, dW_t\right)^2\right] = \int_S^T E[g_t^2] \, dt$
(iii) $\int_S^T g_t \, dW_t$ *is $\mathcal{F}_T^W$-measurable.*

Proof. This will be shown for an elementary function ϕ. From propositions 2.3, 2.4, and 2.5 then follows that it holds for any function $g \in \pounds^2$.
(i)

$$\begin{aligned} E\left[\int_S^T \phi(t,\omega) \, dW_t\right] &= E\left[\sum_{j\geq 0} \phi(t_j,\omega)(W_{t_{j+1}} - W_{t_j})\right] \\ &= \sum_{j\geq 0} E\left[\phi(t_j,\omega)(W_{t_{j+1}} - W_{t_j})\right]. \end{aligned}$$

By definition, the Brownian motion has independent increments, i.e. a forward increment $(W_{t_{j+1}} - W_{t_j})$ is independent of $\phi(t_j,\omega)$. And we have

$$E\left[\phi(t_j,\omega)(W_{t_{j+1}} - W_{t_j})\right] = E[\phi(t_j,\omega)] \cdot \mathbb{N}[W_{t_{j+1}} - W_{t_j}] = 0.$$

□

(ii) This result is known as Itô-Isometry, see Lemma 3.1.5 in [10]. Put $\Delta W_j = W_{t_{j+1}} - W_{t_j}$. Then

$$E[e_i e_j \Delta W_i W_j] = \begin{cases} 0 & \text{if } i \neq j \\ E[e_j^2] \cdot (t_{j+1} - t_j) & \text{if } i = j \end{cases}$$

given the defining properties of the Brownian motion. Thus

$$\begin{aligned} E\left[\left(\int_S^T \phi(t,\omega) \, dW_t(\omega)\right)^2\right] &= \sum_{i,j} E[e_i e_j \Delta W_i \Delta W_j] = \sum_j E[e_j^2] \cdot (t_{j+1} - t_j) \\ &= E\left[\int_S^T \phi(t,\omega)^2 \, dt\right] \end{aligned}$$

□

(iii) Follows from the fact both $\phi(t,\omega)$ and W_t are adapted to the $\mathcal{F}_t^W$-filtration.

2.3. **Stochastic Differential Equations.** Differential equations are used in most parts of science to explain and model movements over time using their momentary velocities. Deterministic differential equations gives deterministic models whereas

stochastic differential equations returns a stochastic process as a model. Consider the stochastic equation

$$dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dW_t$$

where μ and σ are either deterministic functions depending on (t, x) or random coefficients depending on $(t, x, \alpha_t(\omega))$ where α_t is a progressively measurable process.

We say that a solution X to the stochastic differential equation above is strong if it is progressively measurable and the following relation holds true a.s.

$$X_s = X_t + \int_t^s \mu(u, X_u)du + \int_t^s \sigma(u, X_u)dW_u, \quad t \leq s \in [0, T]$$

and

$$\int_t^s |\mu(u, X_u)|du + \int_t^s |\sigma(u, X_u)|^2 du < \infty.$$

Strong solutions require pathwise uniqueness i.e. depending on the filtration, all solutions are almost surely indistinguishable. To ensure the existence and uniqueness of a strong solution the following conditions must be met:

- There exists a deterministic constant K such that

$$|\mu(t, x, \omega) - \mu(t, y, \omega)| + |\sigma(t, x, \omega) - \sigma(t, y, \omega)| \leq K|x - y|.$$

- Additionally there exists a real-valued process κ such that

$$|\mu(t, x, \omega)| + |\sigma(t, x, \omega)| \leq \kappa_t(\omega) + K|x|$$

$$E\left[\int_0^t |\kappa_u|^2 du\right] < \infty.$$

These conditions can be found in Section 1.3.1 in [12].

2.4. **Itô's Lemma.** With the definition of the Itô-integral and stochastic differential equations in place it is time to look closer at stochastic differentials. In deterministic calculus we use the **chain rule**

$$\frac{dg}{dt} = \frac{df}{dx}\frac{dx}{dt}.$$

to differentiate a function g w.r.t. the variable t when the functions f, x and g are given by

$$g = f(x(t)), \quad f, x \in C^1.$$

In this thesis a tool to differentiate a stochastic process Z_t, that is a transformation of a stochastic process X_t, is needed. To this end, we will use **Itô's Lemma**.

Theorem 2.8 (Theorem 4.11 in [2])**.** *Assume that the process X has a stochastic differential given by*

$$dX_t = \mu_t\, dt + \sigma_t\, dW_t$$

where μ and σ are adapted processes, and let f be a $C^{1,2}$-function. Define the process Z by $Z_t = f(t, X_t)$. Then Z has a stochastic differential given by

$$dZ_t = \left(\frac{\partial f}{\partial t}(t, X_t) + \mu_t \frac{\partial f}{\partial x}(t, X_t) + \frac{\sigma_t^2}{2}\frac{\partial^2 f}{\partial x^2}(t, X_t)\right) dt + \sigma_t \frac{\partial f}{\partial x}(t, X_t)dW_t.$$

The term

$$\mu \frac{\partial f}{\partial x} + \frac{\sigma^2}{2} \frac{\partial^2 f}{\partial x^2}$$

will often be referred to as the infinitesimal generator of the underlying process, X, and is a differential operator which in this case is applied to the function F.
A full proof of Itô's Lemma is rather technical and since it is not at the center of this thesis, we will settle for the idea behind the proof. The interested reader is referred to Itô's original article [3] or for a more modern reference see Theorem 4.1.2 in [10]. Below is an intuitive sketch of the idea behind the proof. The reasoning follows the idea in [14].

Proof. Let X_t and Z_t be defined as above. Then a Taylor-expansion of $f(t + dt, X_t + dX_t)$ will be

$$f(t+\Delta t, X_t+\Delta X_t) = f(t, X_t)+\frac{\partial f}{\partial t}\Delta t+\frac{\partial f}{\partial x}\Delta X_t+\frac{1}{2}\frac{\partial^2 f}{\partial x^2}(\Delta X_t)^2+R((\Delta X_t)^3, (\Delta t)^2)$$

and since

$$\Delta X_t = X_{t+\Delta t} - X_t = \mu_t \Delta t + \sigma_t \Delta W_t$$

we see that

$$\begin{aligned} f(t + \Delta t, X_t + \Delta X_t) \approx f(t, X_t) + \frac{\partial f}{\partial t}\Delta t + \frac{\partial f}{\partial x}(\mu_t \Delta t + \sigma_t \Delta W_t) + \\ + \frac{1}{2}\frac{\partial^2 f}{\partial x^2}(\mu_t^2 (\Delta t)^2 + 2\mu_t \sigma_t \Delta t \Delta W_t + \sigma_t^2 (\Delta W_t)^2). \end{aligned} \tag{1}$$

From the definition of the standardised Brownian motion we recall

$$\begin{aligned} E[\Delta W] &= 0 \\ E[(\Delta W)^2] = Var[\Delta W] &= \Delta t \\ Var[(\Delta W)^2] &= 2(\Delta t)^2. \end{aligned}$$

Note that $Var[(\Delta W)^2] \to 0$ at the same time, but faster than $\Delta t \to 0$ therefore we expect $(\Delta W)^2$ to behave more and more deterministically as $\Delta t \to 0$. The informal reasoning above gives us the following values

$$(\Delta t)^2 = 0, \quad \Delta t \Delta W = 0, \quad (\Delta W)^2 = \Delta t$$

to plug into (1)

$$f(t + \Delta t, X + \Delta X_t) \approx f(t, X_t) + \frac{\partial f}{\partial t}\Delta t + \frac{\partial f}{\partial x}(\mu_t \Delta t + \sigma_t \Delta W_t) + \frac{\sigma_t^2}{2}\frac{\partial^2 f}{\partial x^2}\Delta t.$$

Which brings us to the result

$$dZ_t = df(t, X_t) = \left(\frac{\partial f}{\partial t} + \mu_t \frac{\partial f}{\partial x} + \frac{\sigma_t^2}{2}\frac{\partial^2 f}{\partial x^2}\right) dt + \sigma_t \frac{\partial f}{\partial x} dW_t.$$

□

Remark 2.9. *For the multidimensional case: Let X be a n-dimensional process with the dynamics*

$$dX = \mu_t dt + \sigma_t dW_t$$

then the process $Z_t = f(t, X_t)$ has the differential

$$dZ_t = \left(\frac{\partial f}{\partial t} + \sum_{i=1}^{n} \mu_t^i \frac{\partial f}{\partial x^i} + \frac{1}{2} \sum_{i,j=1}^{n} C_t^{i,j} \frac{\partial^2 f}{\partial x^i \partial x^j} \right) dt + \sum_{i=1}^{n} \sigma_t^i \frac{\partial f}{\partial x^i} \, dW_t.$$

The matrix $C = \sigma\sigma^*$ where * denotes transpose.

2.5. **Feynman-Kac Formula.** The Feynman Kac formula is a connection between parabolic partial differential equations and stochastic processes. More specifically, the solutions to certain PDEs are the expected value of some functional depending on a stochastic process X. The idea of combining deterministic PDEs with stochastic processes is later central when we look at the Hamilton-Jacobi-Bellman equation.

Proposition 2.10 (Proposition 5.5 in [2])**.** *Assume that F is a solution to the boundary value problem*

$$\frac{\partial F}{\partial t}(t,x) + \mu(t,x)\frac{\partial F}{\partial x} + \frac{1}{2}\sigma^2(t,x)\frac{\partial^2 F}{\partial x^2} = 0,$$
$$F(T,x) = \Phi(x).$$

Assume furthermore that the process

$$\sigma(s, X_s)\frac{\partial F}{\partial x}(s, X_s)$$

is in $\pounds^2$, where X is defined below. Then F has the representation

$$F(t,x) = E[\Phi(X_T)],$$

where X satisfies the SDE

$$dX_s = \mu(s, X_s)\, ds + \sigma(s, X_s)\, dW_s,$$
$$X_t = x$$

Proof. Consider the PDE and the terminal condition above. Fix a point t in time and a point x in space. Define the stochastic process X on $[t, T]$ as the solution to the SDE

$$dX_s = \mu(s, X_s)\, ds + \sigma(s, X_s)\, dW_s$$
$$X_t = x$$

The infinitesimal generator for this process $\mathcal{L}$ is given by

$$\mathcal{L} = \mu(t,x)\frac{\partial}{\partial x} + \frac{1}{2}\sigma^2(t,x)\frac{\partial^2}{\partial x^2},$$

giving us the PDE above rewritten as

$$\frac{\partial F}{\partial t}(t,x) + \mathcal{L}F(t,x) = 0,$$
$$F(T,x) = \Phi(x).$$

By applying Itô's Lemma to $F(t, X_t)$ between t and T we get

$$F(T, X_T) = F(t, X_t) + \int_t^T \frac{\partial F}{\partial t}(t, x) + \mathcal{L}F(t, x)\, ds + \int_t^T \sigma(s, X_s) \frac{\partial F}{\partial x} dW_s.$$

Since F by assumption satisfies the PDE, the deterministic integral above will vanish.

$$\sigma(s, X_s) \frac{\partial F}{\partial x}(s, X_s)$$

is assumed to be integrable and the expected value of the Itô-integral will therefore vanish. We are now left with

$$E[F(T, X_T) - F(t, X_t)] = 0 \implies E[\Phi(X_T) - F(t, x)] = 0 \implies E[\Phi(X_T)] = F(t, x).$$

□

Remark 2.11. *Note that the process X_t is stochastic and the solution to the PDE is deterministic. What bridges this gap is the expected value. The solution is the case where we instead of looking at each trajectory are looking at the expected value of the stochastic process, which is deterministic.*

3. Stochastic Optimal Control

The basic understanding of stochastic calculus accumulated in the previous section shall now be put to use. We start by defining the problem we seek to solve. The problem is to maximize the **instantaneous utility function** f up to time T combined with the **terminal utility function** g, by applying some control to the stochastic process governing the movements of utility. More often will this be referred to as optimizing the **value function** over the set of admissible controls. To be more precise, we start by defining the value function J given a control α:

Definition 3.1. *The **value function** $J(t, x, \alpha)$ is defined by*

$$J(t, x, \alpha) = E\left[\int_t^T f(s, X_s^{t,x}, \alpha_s)ds + g(X_T^{t,x})\right].$$

$X_s^{t,x}$ is the **state process** with the dynamics

$$dX_t = \mu(t, X_t, \alpha_t)dt + \sigma(t, X_t, \alpha_t)dW_t$$

starting at time t with the start value x, W_t is a standardised Brownian motion. α_t is the **control process** at time t which is used to steer the value function to the optimal value. In most cases we will need to impose some constraints on the control process to keep the problem tractable. Let the fixed subset $A \subseteq \mathbb{R}$ be the set of controls that satisfies these constraints.

Definition 3.2 (Definition 25.1 in [2])**.** *A control process, α, is said to be admissible if*

$$\alpha(t, x) \in A, \quad \forall (t, x) \in [0, \infty) \times \mathbb{R}$$

and for any given initial point (t, x) there exists a unique solution to the SDE

$$dX_s = \mu(s, X_s, \alpha(s, X_s))dt + \sigma(s, X_s, \alpha(s, X_s))dW_t$$
$$X_t = x.$$

The class of admissible controls is denoted by $\mathcal{A}$.

To ensure that every admissible control process α_t is adapted to the $\mathcal{F}_t^W$-filtration we will define it as a function depending on the starting values (t, X_t).

Definition 3.3. *We say that a deterministic function $g(t, x)$ is a **feedback control law** if*

$$\alpha_t = g(t, X_t) \ , \forall \alpha_t \in \mathcal{A}(t, x).$$

Note that the stochastic part of the state process is the Brownian motion which by definition is continuous and adapted to the $\mathcal{F}_t^W$-filtration, hence, if α is a feedback control law then α is adapted to the $\mathcal{F}_t^W$-filtration.

Definition 3.4. *The **optimal value function** $v(t, x)$ is defined as*

$$v(t, x) = J(t, x, \hat{\alpha})$$

*where $\hat{\alpha}$ is the the **optimal control process** such that*

$$v(t, x) = J(t, x, \hat{\alpha}) = \sup_{\alpha \in \mathcal{A}(t,x)} E\left[\int_t^T f(s, X_s^{t,x}, \alpha_s)ds + g(X_T^{t,x})\right]. \tag{2}$$

We are now ready to formulate the problem itself.

Definition 3.5. *The optimal control problem is defined as finding the optimal control process $\hat{\alpha}$ given a value function $J(t,x,\alpha)$, a state process with the dynamics*

$$dX_s^{\alpha} = \mu(s, X^{t,x}, \alpha_s(X^{t,x}))ds + \sigma(s, X_s^{t,x}, \alpha_s(X_s^{t,x}))dW_s,$$

and the class of admissible controls $\mathcal{A}(s,y)$.

Note the notation s, y above is to avoid a conflict with the initial point (t,x).

Remark 3.6. *One might as well consider a minimizing problem by defining the optimal value function and optimal control as*

$$v(t,x) = J(t,x,\hat{\alpha}) = \sup_{\alpha\in\mathcal{A}} E\left[-\left(\int_t^T f(s, X_s^{t,x}, \alpha_s)ds + g(X_T^{t,x})\right)\right].$$

The approach we will take to this problem can be divided into four steps. The first is to formulate the problem, we need to know how the value function is defined and to find an SDE that represents the state process. The second step is to use dynamic programming to divide the problem into sub-problems from which we derive a PDE called a Hamilton-Jacobi-Bellman equation which should hold for any (t,x) in a certain domain given that the optimal value function is inserted. The third step starts with choosing an ansatz, i.e. making a qualified guess of what a solution to the Hamilton-Jacobi-Bellman equation should look like. Differentiate and plug the ansatz into the Hamilton-Jacobi-Bellman equation. Now try to solve the equation. The fourth step is to take the solution of the Hamilton-Jacobi-Bellman equation and verify that this solution is indeed the optimal value function and that it is utilizing the optimal control. This approach is based on the studies published by Bellman in 1954[1].

3.1. **Dynamic Programming Principle.** The Dynamic Programming Principle states that for a controlled Markov process, i.e. a process which is independent of all movements in the past, we can divide the optimization problem into the sub-intervals $[t,\theta], [\theta,T]$. We find the optimal control on $[t,T]$ by searching for an optimal control with the initial point $(\theta, X_\theta^{t,x})$ and then maximizing the expected value over the controls on $[t,\theta]$

$$v(t,x) = \sup_{\alpha\in\mathcal{A}(t,x)} E\left[\int_t^\theta f(s, X_s^{t,x}, \alpha_s)ds + v(\theta, X_\theta^{t,x})\right].$$

This might seem like we have doubled our problem, but the fact is that this shows that the optimal control is independent of the starting time. If we find the optimal control on $[t,\theta]$ we know the optimal control on $[t,T]$.

Theorem 3.7 (Theorem 3.3.1 in [12])**.** *Let $(t,x) \in [0,T]\times\mathbb{R}^n$. Then we have*

$$\begin{aligned} v(t,x) &= \sup_{\alpha\in\mathcal{A}(t,x)} \sup_{\theta\in[t,T]} E\left[\int_t^\theta f(s, X_s^{t,x}, \alpha_s)ds + v(\theta, X_\theta^{t,x})\right] \\ &= \sup_{\alpha\in\mathcal{A}(t,x)} \inf_{\theta\in[t,T]} E\left[\int_t^\theta f(s, X_s^{t,x}, \alpha_s)ds + v(\theta, X_\theta^{t,x})\right] \end{aligned}$$

Proof. This proof will be done in one direction at a time.
(i) For $\alpha \in \mathcal{A}(t,x)$, we have, for any stopping time $\theta \in [t,T]$, the Markovian structure

$$X_s^{t,x} = X_s^{\theta, X_\theta^{t,x}}, \ s \geq \theta.$$

Then by iterated expectations

$$J(t,x,\alpha) = E\left[\int_t^\theta f(s, X_s^{t,x}, \alpha_s)ds + J(\theta, X_\theta^{t,x}, \alpha)\right].$$

By definition the inequality $J(t,x,\alpha) \leq v(t,x)$, is true for all $\theta \in [t,T]$, thus

$$J(t,x,\alpha) \leq \inf_{\theta\in[t,T]} E\left[\int_t^\theta f(s, X_s^{t,x}, \alpha_s)ds + v(\theta, X_\theta^{t,x})\right]. \tag{3}$$

Taking the supremum over $\alpha \in \mathcal{A}(t,x)$ on both sides of inequality (3) we see that

$$v(t,x) \leq \sup_{\alpha\in\mathcal{A}(t,x)} \inf_{\theta\in[t,T]} E\left[\int_t^\theta f(s, X_s^{t,x}, \alpha_s)ds + v(\theta, X_\theta^{t,x})\right]. \tag{4}$$

(ii) Fix some arbitrary control $\alpha \in \mathcal{A}(t,x)$ and $\theta \in [t,T]$. By the definition of the functions $J(t,x,\alpha)$ and $v(t,x)$ there exists, for any $\varepsilon > 0$ and $\omega \in \Omega$, $\alpha^{\varepsilon,\omega} \in \mathcal{A}(\theta(\omega), X_{\theta(\omega)}^{t,x}(\omega))$ such that

$$v(\theta(\omega), X_{\theta(\omega)}^{t,x}(\omega)) - \varepsilon \leq J(\theta(\omega), X_{\theta(\omega)}^{t,x}(\omega), \alpha^{\varepsilon,\omega}).$$

Introduce the process α^* defined as

$$\alpha_s^*(\omega) = \begin{cases} \alpha_s(\omega), & s \in [0, \theta(\omega)] \\ \alpha_s^{\varepsilon,\omega}(\omega), & s \in [\theta(\omega), T]. \end{cases}$$

The process α^* is continuous and thus admissible (this is a deep result as we can not easily guarantee continuity at $\theta(\omega)$, see [12]). By the definition of the functions and through iterated expectations follows

$$v(t,x) \geq J(t,x,\alpha^*) = E\left[\int_t^\theta f(s, X_s^{t,x}, \alpha_s)ds + J(\theta, X_\theta^{t,x}, \alpha_s^\varepsilon)\right]$$

$$\geq E\left[\int_t^\theta f(s, X_s^{t,x}, \alpha_s)ds + v(\theta, X_\theta^{t,x})\right] - \varepsilon.$$

Given that $\alpha \in \mathcal{A}$, $\theta[t,T]$ and $\varepsilon > 0$ was chosen arbitrarily the following holds

$$v(t,x) \geq \sup_{\alpha\in\mathcal{A}(t,x)} \sup_{\theta\in[t,T]} E\left[\int_t^\theta f(s, X_s^{t,x}, \alpha_s)ds + v(\theta, X_\theta^{t,x})\right]. \tag{5}$$

Combining (4) and (5) we see that

$$v(t,x) = \sup_{\alpha\in\mathcal{A}(t,x)} E\left[\int_t^\theta f(s, X_s^{t,x}, \alpha_s)ds + v(\theta, X_\theta^{t,x})\right], \quad \forall\theta \in [t,T].$$

□

3.2. **Hamilton Jacobi Bellman Equation.** Using the dynamic programming principle we will show that the optimal value function is a solution to the partial differential equation known as the Hamilton-Jacobi-Bellman equation, which is referred to as the HJB equation below. To derive the HJB equation we shall look at the behavior of the value function at an arbitrary small interval $[t, t+h]$, and thus fix a point (t, x). The dynamic programming principle ties it together with the value function on the whole interval $[t, T]$. Below follows a formal derivation of the HJB equation.

Derivation 3.8 (Derivation 3.4.1 in [12])**.**

Let $\theta = t + h$ and introduce an arbitrary constant control $a \in \mathcal{A}(t, x)$, and let $\alpha_s = a$. Then from the dynamic programming principle follows

$$v(t,x) \geq E\left[\int_t^{t+h} f(s, X_s^{t,x}, a)ds + v(t+h, X_{t+h}^{t,x})\right]. \tag{6}$$

Assume that $v(t, x) \in C^{1,2}$ and apply Itô's Lemma between t and $t+h$, then follows

$$v(t+h, X_{t+h}^{t,x}) = v(t,x) + \int_t^{t+h} \left(\frac{\partial v}{\partial t} + \mathcal{L}^a v\right)(s, X_s^{t,x})ds + \int_t^{t+h} \frac{\partial v}{\partial x}(s, X_s^{t,x})\sigma^a dW_s,$$

where $\mathcal{L}^a v$ is the infinitesimal generator of the state process given the constant control a

$$\mathcal{L}^a v = \mu(x,a)\frac{\partial v}{\partial x} + \frac{\sigma(x,a)^2}{2}\frac{\partial^2 v}{\partial x^2}.$$

Substituting $v(t+h, X_{t+h}^{t,x})$ in (6), subtracting $v(t, x)$ from both sides and considering that the expected value of the Itô-integral

$$E\left[\int_t^{t+h} \frac{\partial v}{\partial x}(s, X_s^{t,x})\sigma^a dW_s\right] = 0,$$

we get

$$0 \geq E\left[\int_t^{t+h} \left(\frac{\partial v}{\partial t} + \mathcal{L}^a v\right)(s, X_s^{t,x}) + f(s, X_s^{t,x}, a)ds\right].$$

If we divide both sides with h and let $h \to 0$ we get

$$0 \geq E\left[\lim_{h\to 0} \frac{\int_t^{t+h} \left(\frac{\partial v}{\partial t} + \mathcal{L}^a v\right)(s, X_s^{t,x}) + f(s, X_s^{t,x}, a)ds}{h}\right],$$

from the Mean Value Theorem then follows

$$0 \geq \frac{\partial v}{\partial t}(t,x) + \mathcal{L}^a v(t,x) + f(t,x,a).$$

And since $a \in A$ was chosen arbitrary, the following must hold

$$-\frac{\partial v}{\partial t}(t,x) - \sup_{a\in A}[\mathcal{L}^a v(t,x) + f(t,x,a)] \geq 0.$$

If we choose the optimal control $\hat{\alpha} \in \mathcal{A}(t, x)$ instead of a, (6) would by definition be the equation

$$v(t,x) = E\left[\int_t^{t+h} f(s, X_s^{t,x}, \hat{\alpha})ds + v(t+h, X_{t+h}^{t,x})\right].$$

With analogous reasoning as above, we see that

$$-\frac{\partial v}{\partial t}(t,x) - \mathcal{L}^{\hat{\alpha}} v(t,x) - f(t,x,\hat{\alpha}) = 0.$$

And since $\hat{\alpha}$ is defined by

$$\hat{\alpha} = \arg\max_{\alpha \in \mathcal{A}} J(t,x,\alpha)$$

the following should hold

$$-\frac{\partial v}{\partial t}(t,x) - \sup_{\alpha \in \mathcal{A}} [\mathcal{L}^{\alpha} v(t,x) + f(t,x,\alpha)] = 0 \ \ , \forall (t,x) \in [0,T) \times \mathbb{R}.$$

This is the Hamilton-Jacobi-Bellman equation [12]. The terminal condition associated to the HJB equation,

$$v(T,x) = g(x) \ \ , \forall x \in \mathbb{R},$$

is recognisable from (2). The function $\sup_{\alpha \in \mathcal{A}} [\mathcal{L}^{\alpha} v(t,x) + f(t,x,\alpha)]$ is called the Hamiltonian of the control problem. We now summarize the HJB equation as

$$\begin{cases} \frac{\partial v}{\partial t} + \sup_{a \in A} \left[\mu(x,a) \frac{\partial v}{\partial x} + \frac{\sigma(x,a)^2}{2} \frac{\partial^2 f}{\partial x^2} + f(t,x,a) \right] = 0, \quad \forall (t,x) \in [0,T) \times \mathbb{R} \\ v(T,x) = g(x) \ \ , \forall x \in \mathbb{R}. \end{cases}$$

Remark 3.9. *In the case where the class of admissible controls consists of only a single control, $\mathcal{A} = \{\alpha_0\}$, there is no longer an optimization problem and the corresponding PDE is solvable through the Feynman Kac formula, section 2.5 above, given a known state process X.*

3.3. **Verification Theorem.** In the previous section, the HJB equation was derived. It was stated that the optimal value function solves the PDE, but we can not yet guarantee that a solution to the HJB equation is the optimal value function. This is where the Verification Theorem comes in. It states that any solution to the HJB equation is indeed the optimal value function.

Theorem 3.10 (Theorem 3.5.2 in [12])**.** *Let $w(t,x)$ be a function in $C^{1,2}([0,T) \times \mathbb{R}^n) \cap C^0([0,T] \times \mathbb{R}^n)$, and satisfying a quadratic growth condition, i.e. there exists a constant C such that*

$$|w(t,x)| \leq C(1 + |x|^2), \quad \forall (t,x) \in [0,T] \times \mathbb{R}^n.$$

(i) *Suppose that*

$$-\frac{\partial w}{\partial t}(t,x) - \sup_{a \in A} [\mathcal{L}^a w(t,x) + f(t,x,a)] \geq 0, \quad (t,x) \in [0,T) \times \mathbb{R}^n \tag{7}$$

$$w(T,x)) \geq g(x), \quad x \in \mathbb{R}^n$$

then $w(t,x) \geq v(t,x)$, on $[0,T] \times \mathbb{R}^n$.

(ii)*Suppose further that $w(T) = g$, and that there exists a measurable function $\hat{\alpha}(t,x)$, $(t,x) \in [0,T] \times \mathbb{R}^n$, valued in A such that*

$$-\frac{\partial w}{\partial t}(t,x) - \sup_{a \in A} [\mathcal{L}^a w(t,x) + f(t,x,a)] = -\frac{\partial w}{\partial t}(t,x) - \mathcal{L}^{\hat{\alpha}} w(t,x) - f(t,x,\hat{\alpha}(t,x)) = 0,$$

the SDE

$$dX_s = \mu(X_s, \hat{\alpha}(s,X_s))ds + \sigma(X_s, \hat{\alpha}(s,X_s))dW_s$$

admits a unique solution, denoted by $\hat{X}_s^{t,x}$, given an initial condition $X_t = x$ and the process $\{\hat{\alpha}(s, \hat{X}_s^{t,x})t \leq s \leq T\}$ lies in $\mathcal{A}(t,x)$. Then

$$w = v \quad on\ [0,T] \times \mathbb{R}^n,$$

and $\hat{\alpha}$ is an optimal Markovian control.

Proof. ***(i)*** Given $w \in C^{1,2}$, by Itô's Lemma, the following holds, for all $(t,x) \in [0,T] \times \mathbb{R}^n, \alpha \in \mathcal{A}(t,x), s \in [t,T]$ and any stopping time $\tau \in [t,\infty)$

$$w(s \wedge \tau, X_{s\wedge\tau}^{t,x}) = w(t,x) + \int_t^{s\wedge\tau} \frac{\partial w}{\partial t}(u, X_u^{t,x}) + \mathcal{L}^{\alpha_u} w(u, X_u^{t,x}) du$$
$$+ \int_t^{s\wedge\tau} \frac{\partial w}{\partial x}(u, X_u^{t,x}) \sigma(X_u^{t,x}, \alpha_u) dW_u.$$

Choose $\tau = \tau_n = \inf\{s \geq t : \int_t^s |\frac{\partial w}{\partial x}(u, X_u^{t,x})\sigma(X_u^{t,x}, \alpha_u|^2 du \geq n\}$. Notice that $\tau_n \to \infty$ as $n \to \infty$.
The stopped process

$$\int_t^s \frac{\partial w}{\partial x}(u, X_u^{t,x}) \sigma(X_u^{t,x}, \alpha_u) dW_u, \quad s \in [t,T],$$

is a martingale and thus,

$$E[w(s \wedge \tau, X_{s\wedge\tau}^{t,x})] = w(t,x) + E\left[\int_t^{s\wedge\tau} \frac{\partial w}{\partial t}(u, X_u^{t,x}) + \mathcal{L}^{\alpha_u} w(u, X_u^{t,x} du\right].$$

By assumption (7), we have

$$\frac{\partial w}{\partial t}(u, X_u^{t,x}) + \mathcal{L}^{\alpha_u} w(u, X_u^{t,x}) + f(X_u^{t,x}, \alpha_u) \leq 0 \ , \forall \alpha \in \mathcal{A}(t,x)$$

and so

$$E[w(s \wedge \tau, X_{s\wedge\tau}^{t,x})] \leq w(t,x) + E\left[\int_t^{s\wedge\tau} f(X_u^{t,x}, \alpha_u) du\right]. \tag{8}$$

From the triangle inequality we get

$$\left|\int_t^{s\wedge\tau_n} f(X_u^{t,x}, \alpha_u) du\right| \leq \int_t^{s\wedge\tau_n} |f(X_u^{t,x}, \alpha_u)| du.$$

By the integrability condition in the construction of $\mathcal{A}(t,x)$ we know hat the right hand term is integrable. And since w by assumption satisfies a quadratic growth condition;

$$|w(s \wedge \tau_n, X_{s\wedge\tau_n}^{t,x})| \leq k(1 + \sup_{s\in[t,T]} |X_s^{t,x}|^2)$$

and we know that $k(1 + \sup_{s\in[t,T]} |X_s^{t,x}|^2)$ is integrable. The Dominated Convergence Theorem, see Theorem 11.32 in [13], can be applied and shows that when sending $n \to \infty$ in (8), $|w(s \wedge \tau_n, X_{s\wedge\tau_n}^{t,x})|$ is also integrable. In the definition of τ_n we see that $s \wedge \tau_n \to s$ as $n \to \infty$. Let $n \to \infty$, then

$$E[w(s, X_s^{t,x})] \leq w(t,x) - E\left[\int_t^s f(X_u^{t,x}, \alpha_u)\right] \ , \forall \alpha \in \mathcal{A}(t,x). \tag{9}$$

The fact that w is continuous on $[0,T] \times \mathbb{R}^n$ lets us send $s \to T$, remembering $w(T,x) \geq g(x)$ gives us from (9)

$$E[g(X_T^{t,x})] \leq w(t,x) - E\left[\int_t^T f(X_u^{t,x}, \alpha_u) du\right] \ , \forall \alpha \in \mathcal{A}(t,x)$$

and thus

$$J(t,x,\alpha) \leq w(t,x) \ \ ,\forall \alpha \in \mathcal{A}(t,x).$$

Through the arbitrariness of α follows

$$w(t,x) \geq v(t,x).$$

(ii) Apply Itô's Lemma to $w(u, \hat{X}_u^{t,x})$ between t and s where $t, s \in [0,T), s \geq t$. Then follows (after an eventual localisation for removing the Itô-integral term in the expectation),

$$E\left[w(s, \hat{X}_s^{t,x})\right] = w(t,x) + E\left[\int_t^s \frac{\partial w}{\partial t}(u, \hat{X}_u^{t,x}) + \mathcal{L}^{\hat{\alpha}(u,\hat{X}_u^{t,x})} w(u, \hat{X}_u^{t,x}) du\right].$$

From the definition of $\hat{\alpha}(t,x)$ follows

$$-\frac{\partial w}{\partial t} - \mathcal{L}^{\hat{\alpha}(t,x)} w(t,x) - f(t,x,\hat{\alpha}(t,x)) = 0,$$

and thus

$$E\left[w(s, \hat{X}_s^{t,x})\right] = w(t,x) - E\left[\int_t^s f(\hat{X}^t t, x_u, \hat{\alpha}(u, \hat{X}_u^{t,x})) du\right].$$

We see that sending $s \to T$ gives

$$w(t,x) = E\left[\int_t^T f(\hat{X}^t t, x_u, \hat{\alpha}(u, \hat{X}_u^{t,x})) du + g(\hat{X}_T^{t,x})\right] = J(t,x,\hat{\alpha}).$$

To conclude the proof we notice that $w(t,x) = J(t,x,\hat{\alpha}) \leq v(t,x)$ and thus, $w(t,x) = v(t,x)$ with $\hat{\alpha}$ as an optimal control.

$\square$

4. Merton's Portfolio Problem

4.1. **Introduction to the Problem.** Before looking at Merton's portfolio problem as a whole where we consider both the utility of consumption and terminal wealth, we will divide it into two partial problems and look at them separately so that we later can see how the combined problem differs from the two separate ones. The first problem to consider is maximizing the terminal value of wealth at a given time T, e.g. investing in a pension portfolio before retiring at time T. The optimal value function is then defined as

$$v_1(t,x) = \sup_w \left\{ E\left[X^p_{1_T} \right] \right\}$$

with the state process of wealth given from the Black Scholes' market model with one asset including risk, S_t, and one without risk, B_t. Their dynamics are given by

$$dB_t = rB_t\, dt$$

$$dS_t = \mu S_t\, dt + \sigma S_t\, dW_t.$$

The process representing the risky asset is known as a **Geometric Brownian Motion** and has the explicit representation

$$S_t = S_0\, exp\left\{ \left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t \right\}$$

with the explicit expected value and variance

$$E[S_t] = S_0\, exp\{\mu t\}$$
$$Var[S_t] = S_0^2\, exp\{2\mu t\}(exp\{\sigma^2 t\} - 1).$$

We introduce the total wealth X_{1_t} and define it as

$$X_{1_t} = w_S S_t + w_B B_t,$$

where $w_S = \frac{S_t}{X_{1_t}}$ is the relative weight of the risky asset and $w_B = \frac{B_t}{X_{1_t}}$ is the relative weight of the risk free asset. Let $w = w_S$, then we see that it makes sense to define the relative weight of the risk free asset as $(1-w) = w_B$. Note that it is allowed to short sell the risky asset to place a larger weight in the risk free asset as well as borrow money to place a larger weight in the risky assets. This means in practice that there are no constraints on w more that the sum of the relative weights for the risky and the risk free asset equals 1. We now have the dynamics of the total wealth given by

$$dX_{1_t} = w_t(\mu - r)X_{1_t}dt + rX_{1_t}dt + w_t\sigma X_t dW_{1_t},\ \ X_{1_0} = x_1 > 0. \tag{10}$$

There is one control process, the relative weight of the risky asset denoted by w. This is a problem of portfolio allocation with a known stopping time T.

The second part of Merton's portfolio problem is that of maximizing utilization of consumption $c(t, X_t)$ up to time T given the optimal value function

$$v_2(t,x) = \sup_{c,w} \left\{ E\left[\int_0^T exp\{-\beta t\} c_t^p dt \right] \right\},\ \ 0 < p < 1,\ \ 0 \le c.$$

In this case we do not care about the wealth left at time T. The state process representing wealth is still under the Black-Scholes' market model and similar to (10), but with a consumption rate subtracted in the deterministic integral, i.e.

$$dX_{2_t} = w_t(\mu - r)X_{2_t}dt + (rX_{2_t} - c_t)dt + w_t\sigma X_t dW_{2_t}, \;\; X_{2_0} = x_2 > 0. \tag{11}$$

There are two control processes in this case, the consumption rate, denoted by c_t, and the relative weight of the risky asset denoted by w. The constraint on consumption is that it is non-negative, i.e. we can not inject capital. The relative weights of the portfolio have, as in the first part, no constraints. Note that this problem has a subjective discounting rate β, this rate is part of the model and lets the investor differ her valuation of consumption now and in the future. This problem is called an investment/consumption was originally studied by Merton in [6], and then extended to more general value functions and other price behaviour assumptions by Merton in [7].

After formulating each problem separately it is now time to formulate Merton's portfolio problem. How does an investor maximize utilization of both consumption up to time T and the terminal value of wealth at time T? For the same modeling reasons as above a subjective discounting factor β, is included both in regards to the consumption and the terminal wealth.

Define the class of admissible controls $\mathcal{C} \times \mathcal{W}$ as

$$\mathcal{C} = \{c(t,x) : c(t,x) \in [0,\infty), \;\; \forall (t,x) \in [0,T] \times [0,\infty)\}$$

$$\mathcal{W} = \{w(t,x) : w(t,x) \in \mathbb{R}, \;\; \forall (t,x) \in [0,T] \times [0,\infty)\}.$$

This leaves us with the problem formulation

$$\sup_{(c,w)\in\mathcal{C}\times\mathcal{W}} \left\{ E\left[\int_0^T exp\{-\beta t\} c_t^p \, dt + exp\{-\beta t\} X_T^p \right] \right\}, \;\; 0 < p < 1,$$

where the wealth process X_t once again is given under the Black-Scholes' market model with consumption as in (11).

$$dX_t = w_t(\mu - r)X_t dt + (rX_t - c_t)dt + w_t\sigma X_t dW_t, \;\; X_0 = x > 0.$$

w_t is the relative weight of the risky asset and c_t is the consumption rate.
We consider both the modelling parameters β, and p, as well as the market parameters μ, r, and σ to be known positive constants. Before we dig in to the problem we should summarize from intuition what we expect of the solution:

- $v(t,x)$ is real valued, something else would not make sense,
- $c \geq 0$, a negative consumption is not allowed.

Remark 4.1. *In Merton's original version of the problem a weight ϵ^{1-p}, $0 \leq \epsilon \leq 1$ is multiplied with the bequest function as a way of further modeling the difference in the value of wealth consumed and wealth left as an inheritance. This will slightly change the optimal value function and optimal consumption rate below by a constant term.*

4.2. **Solving the problem.** As seen in sections 3.2 and 3.3, solving this problem is equivalent to finding a solution to the corresponding Hamilton Jacobi Bellman equation

(12)

$$\frac{\partial v}{\partial t} + \sup_{(c,w)\in\mathcal{C}\times\mathcal{W}} \left\{ exp\{-\beta t\}c^p + wx(\mu - r)\frac{\partial v}{\partial x} + (rx - c)\frac{\partial v}{\partial x} + \frac{1}{2}w^2\sigma^2x^2\frac{\partial^2 v}{\partial x^2} \right\} = 0$$

(13)

$$v(T, x) = exp\{-\beta T\}x^p$$

(14)

$$v(t, 0) = 0$$

The boundary condition (14) means that all activity stops when the wealth is 0. Differentiating the Hamiltonian with regard to c and w respectively gives the supremum over each control by

$$p(\hat{c})^{p-1} = exp\{\beta t\}\frac{\partial v}{\partial x}$$

$$\hat{w} = -\frac{\partial v}{\partial x}\left(x\frac{\partial^2 v}{\partial x^2}\right)^{-1}\frac{\mu - r}{\sigma^2}$$

To solve the PDE we make the ansatz that

$$v^*(t, x) = exp\{-\beta t\}f(t)x^p,$$

is the candidate solution. Note that $v^*(t, x)$ is similar to the utility function with an added time dependent factor. We see that

$$\frac{\partial v^*}{\partial t} = exp\{-\beta t\}\frac{df}{dt}x^p - \beta\, exp\{-\beta t\}f(t)x^p,$$

$$\frac{\partial v^*}{\partial x} = p\, exp\{-\beta t\}f(t)x^{p-1},$$

$$\frac{\partial^2 v^*}{\partial x^2} = p(p-1)\, exp\{-\beta t\}f(t)x^{p-2}.$$

And this gives us the candidate optimal controls

$$\hat{c}^* = xf^{-1/(1-p)}$$

$$\hat{w}^* = -\frac{\mu - r}{\sigma^2(p-1)}.$$

Plugging this into the HJB-equation gives

$$exp\{-\beta t\}\frac{df}{dt}x^p - \beta\, exp\{-\beta t\}x^p f(t) + exp\{-\beta t\}x^p f(t)^{-p/(1-p)} - exp\{-\beta t\}x^p f(t)p(\frac{(\mu - r)^2}{\sigma^2(p-1)} +$$

$$+exp\{-\beta t\}x^p r f(t)p - exp\{-\beta t\}x^p f(t)^{-1/(1-p}f(t)p + exp\{-\beta t\}x^p\frac{1}{2}f(t)p\frac{(\mu - r)^2}{\sigma^2(p-1)} = 0.$$

Rearranging the terms and factoring out $exp\{-\beta t\}x^p$

$$exp\{-\beta t\}x^p\left(\frac{df}{dt} + f(t)\left(\frac{p(\mu - r)^2}{2\sigma^2(1-p)} + rp - \beta\right) + f(t)^{-p/(1-p)}(1-p)\right) = 0.$$

Let $A = \frac{p(\mu-r)^2}{2\sigma^2(1-p)} + rp - \beta$. If the HJB equation is to hold for all x and t the following ODE must hold

$$\frac{df}{dt} + Af(t) + (1-p)f(t)^{-p/(1-p)} = 0.$$

This is known as a Bernoulli equation and can be solved explicitly as shown in [11].

Lemma 4.2. *An equation on the form*

$$\frac{df}{dx} + P(x)f = Q(x)f^n$$

has the solution

$$f = \left(I^{-1}\int I(1-n)Q(x)\,dx\right)^{1/(1-n)}.$$

Proof. Multiplying both sides with $(1-n)f^{-n}$ gives

$$(1-n)f^{-n}\frac{df}{dx} + (1-n)P(x)f^{1-n} = (1-n)Q(x).$$

Note that by using the chain rule we get

$$\frac{d}{dx}f^{1-n} = (1-n)f^{-n}\frac{df}{dx}.$$

Then by substituting $h = f^{1-n}$ follows the first order, linear equation

$$\frac{dh}{dx} + (1-n)P(x)h = (1-n)Q(x).$$

Now by multiplying with an integrating factor $I(x) = exp\{\int(1-n)P(x)\,dx\}$

$$I\frac{dh}{dx} + \frac{dI}{dx}h = I(1-n)Q(x)$$

the left hand side is the derivative of the product Ih with regard to x and thus

$$f = \left(I^{-1}\int I(1-n)Q(x)\,dx\right)^{1/(1-n)}.$$

□

Let $h(t) = f(t)^{1/1-p}$ and we get the solution

$$h(t) = k\,exp\{-At/(1-p)\} - \frac{1-p}{A}, \quad A \neq 0.$$

Considering the terminal boundary condition (13) we see that

$$v(T,x) = 1 \implies h(T) = 1 \implies k\,exp\left\{\frac{-AT}{1-p}\right\} - \frac{1-p}{A} = 1.$$

Solving for k

$$k = \left(1 + \frac{1-p}{A}\right)exp\left\{\frac{AT}{1-p}\right\}$$

and thus,

$$h(t) = \left(1 + \frac{1-p}{A}\right)exp\left\{\frac{A(T-t)}{1-p}\right\} - \frac{1-p}{A}.$$

To ensure that the consumption rate is non-negative we need

$$h(t) > 0, \quad \forall t \in [0,T] \tag{15}$$

and by considering each of the following cases; $\frac{1-p}{A} > 0, -1 < \frac{1-p}{A} < 0, \frac{1-p}{A} = -1$, and $\frac{1-p}{A} < -1$, we see that (15) holds. Note that $\frac{1-p}{A} = 0$ does not exist with the constraints on p.

$$f(t) = (h(t))^{(1-p)}$$

$$v^*(t,x) = exp\{-\beta t\}x^p\left(\left(1 + \frac{1-p}{A}\right)exp\left\{\frac{A(T-t)}{1-p}\right\} - \frac{1-p}{A}\right)^{(1-p)}.$$

Straight forward calculations now show that $v^*(t,x)$ and $\hat{c}^*, \hat{w}^*$ solves the HJB equation and thereby satisfies Theorem 3.10, thus follows, $v^*(t,x) = v(t,x)$ and $\hat{c}, \hat{w}$ are optimal controls. We see that the optimal controls are

$$\hat{c} = exp\{-\beta t\}x^p \left(\left(1 + \frac{1-p}{A}\right) exp\left\{\frac{A(T-t)}{1-p}\right\} - \frac{1-p}{A}\right)^{(1-p)},$$

$$\hat{w} = \frac{\mu - r}{\sigma^2(1-p)}. \tag{16}$$

For further references when examine the effects of the parameters, we will use the slightly cluttered, but more expressive formulation of the optimal consumption rate (17)

$$\hat{c}(t,x) = x\left(\left(1 + \frac{1-p}{\frac{p(\mu-r)^2}{2\sigma^2(1-p)} + rp - \beta}\right) exp\left\{\left(\frac{p(\mu-r)^2}{2\sigma^2(1-p)} + rp - \beta\right)(T-t)/(1-p)\right\} - \frac{1-p}{\frac{p(\mu-r)^2}{2\sigma^2(1-p)} + rp - \beta}\right)^{(-1)}.$$

4.2.1. *The Infinite Time Setting.* Before exploring the meaning of these results we shall take a brief look at what happens in the case of an infinite time horizon. We start by investigating what happens to the results from previous section on the limit as T approaches ∞ we see from (15), (17) and (16) that

$$\lim_{T\to\infty} \hat{c}(t,x) = x\left(-\frac{A}{1-p}\right) = x\left(-\frac{p(\mu-r)^2}{2\sigma^2(1-p)^2} - \frac{rp}{1-p} + \frac{\beta}{1-p}\right), \tag{18}$$

$$\lim_{T\to\infty} \hat{w}(t,x) = \hat{w}(t,x).$$

After a more rigorous derivation of the optimal controls in the true infinite time setting, where he solves a solution to the HJB equation without time dependence, Merton concludes in his 1969 article [6] that

$$\hat{c}_\infty(x) := \lim_{T\to\infty} \hat{c}(t,x),$$

$$\hat{w}_\infty := \lim_{T\to\infty} \hat{w}(t,x).$$

Where $\hat{c}_\infty$ and $\hat{w}_\infty$ are the optimal controls given the problem with an infinite time horizon. These results will make it easier to understand the effects of the parameters, even in the finite time setting. To ensure that $v(t,x)$ is finite even in infinite time we need to make sure that the discounting factor $exp\{-\beta t\}$ is strong enough to counter the dynamics of the composite portfolio, which from being a geometric Brownian motion we know has exponential drift. We will therefore impose an additional condition in the infinite time setting

$$\beta > p\left(\frac{(\mu-r)^2(2-p)}{2\sigma^2(1-p)} + r\right) \implies A < 0.$$

A detailed derivation of this condition can be found on page 252 in [6]. In particular this condition ensures a positive consumption rate, that the exponential term in $\hat{c}(t,x)$ goes to 0 as T goes to ∞, and

$$\lim_{t\to\infty} E[v(t,X_t)] = 0.$$

5. Economic Interpretations of Merton's Portfolio Problem

5.1. **Dependence of Time and Wealth.** From the optimal controls of the problem, we see that the optimal consumption rate is linear with regard to wealth. For any given wealth, the same proportion of wealth should be consumed for maximal utilization. Note that as the consumption gets big enough, the wealth gets smaller and as wealth goes to 0 the consumption rate will go to 0, and thus wealth is strictly non-negative.

We see that at time T the optimal consumption rate converges to X_T, the wealth. The momentary consumption rate can exceed the wealth and we can therefore not say from what direction the consumption rate will converge to X_T. Note however that the parameters governing whether $\frac{\partial \hat{c}}{\partial t}$ is positive or negative are constant and thus $\frac{\partial \hat{c}}{\partial t}$ is either strictly positive or strictly negative for all $t \in [0, T]$.

The relative weights of the portfolio are independent of both wealth and time. This is quite remarkable since it means that no matter the investor's wealth or how long the time horizon is, the investor should have constant relative weights between risky and risk-free assets in their portfolio. Note that $\hat{w}$ is the inverse of the Arrow-Pratt measure of relative risk aversion multiplied by $\frac{\mu - r}{\sigma^2}$, for a deeper understanding of risk aversion measurements see [8]. In other words, the relative weight of the risky asset is proportional to the market parameters and inversely proportional to the relative risk aversion. The utility function in question is often referred to as a Constant Relative Risk Aversion (CRRA) utility function and it is thus expected for the relative risk aversion measurement to be constant. A change in the parameters μ, r, σ, and p will however change the relative weights of the portfolio, but they are by construction constant in this specific problem. Since we know that the relative weights are constant we can now express the dynamics of the optimal portfolio as one asset, call it Y, with the dynamics

$$dY_t = Y_t \mu_Y dt + Y_t \sigma_Y dW_t, \quad Y_0 = x.$$

Where μ_Y and σ_Y are the drift and volatility of the optimal portfolio given by

$$\mu_Y = \hat{w}\mu + (1 - \hat{w})r = \frac{(\mu - r)^2}{\sigma^2(1 - p} + r$$

$$\sigma_Y^2 = \hat{w}^2 \sigma^2 = \frac{(\mu - r)^2}{\sigma^2(1 - p)^2}.$$

5.2. **Changes in the Parameters.** Changing the parameters p corresponds to changing the relative risk aversion in the model and we see that the relative weight of the risky asset goes up as p goes up, this is shown by

$$\frac{\partial \hat{w}}{\partial p} = \frac{\mu - r}{\sigma^2(1 - p)^2}.$$

which is positive for all $\mu > r$. In the case $\mu < r$ we see that $\hat{w}$ is negative and, an increase of p implies a further decrease in $\hat{w}$. This translates to increasing the short position in the risky asset leading to bigger risk exposure to risk. This is the intuition behind a negative derivative in the case of $\mu < r$.

To ease our calculations we shall consider the case with an infinite time horizon, note that this will get rid of the time dependence. As stated by Merton in Remark 11 of [6]

> "Because this section is concerned with the qualitative changes in the solution with respect to shifts in the parameters, the more simple form of the infinite time horizon case is examined. The essential difference between $C^*_\infty(t)$ and $C^*(t)$ is the explicit time dependence of $C^*(t)$ which was discussed in section V."

It is sufficient to explore how changed parameters in the infinite setting effects the optimal consumption rate to understand their effect in the finite case. In the infinite time setting, (18) gives the rather simple derivative of $\hat{c}_\infty$ w.r.t p,

$$\frac{\partial}{\partial p} A = \frac{(\mu - r)^2}{2\sigma^2(1-p)^2} > 0$$

for $0 < p < 1$, thus follows

$$\frac{\partial}{\partial p} \frac{A}{1-p} > 0$$

and by multiplying with $-x$

$$\frac{\partial}{\partial p}\left[-x\left(\frac{A}{1-p}\right)\right] < 0 \iff \frac{\partial \hat{c}_\infty}{\partial p} < 0.$$

We see that an increase of p always gives a decrease in consumption and that the opposite holds as well.

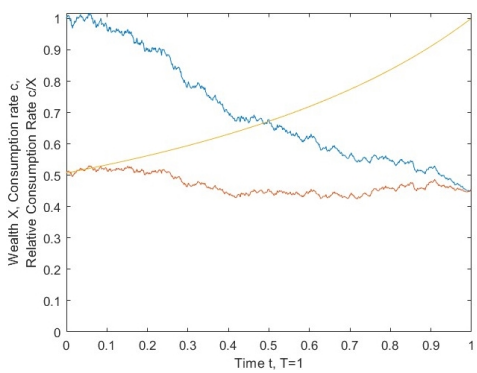

FIGURE 1. $p = 0.1$

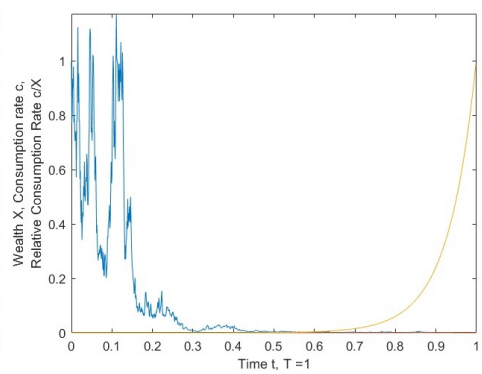

FIGURE 2. $p = 0.97$

Simulations of the problem with high and low risk aversion
$\beta = 0.03,\ \mu = 0.06,\ \sigma = 0.25,\ r = 0.02$

Figure 1 and Figure 2 show simulations in finite time, where the blue line is wealth, the read line is consumption rate, and the yellow line is relative consumption rate, the same coloring convention is used throughout the rest of the figures. The parameters μ and σ are chosen to be similar to OMXS30 during the past 20 years [9] and r is close to the current yield on Swedish 10 year treasury bonds [4] (2022-06-09). The modeling parameters β and p are subjective and can be changed for each investor. We see how a higher p in Figure 2, i.e. lower risk aversion, gives a lower relative consumption rate and a higher exposure to the risky asset. Note that while this is a simulation of a stochastic event, the relative consumption rate is deterministic and therefore does not change between simulations. The effect of

the subjective discount rate β comes rather straight forward from looking at the derivative

$$\frac{\partial \hat{c}_\infty}{\partial \beta} = \frac{x}{1-p}$$

and remembering our assumptions on x and p we see that this is positive. A higher value of β translates to a bigger value depreciation over time and a higher consumption rate can be said to secure the present value. This can be seen in Figure 3 and Figure 4 below.

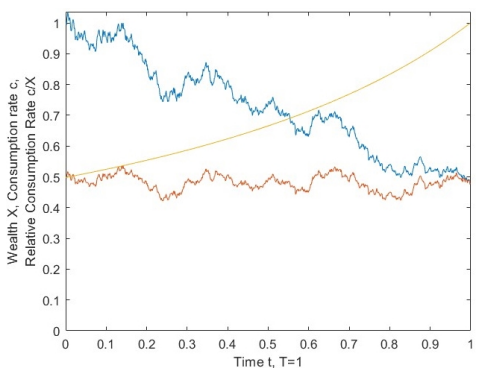

FIGURE 3. $\beta = 0.02$.

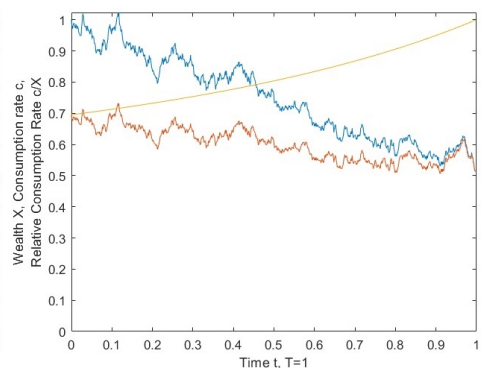

FIGURE 4. $\beta = 0.25$.

Simulations of the problem with low and high subjective discount rate
$p = 0.5,\ \mu = 0.06,\ \sigma = 0.25,\ r = 0.02$

The effects of the market parameters on optimal relative weights and optimal consumption rate can easily be seen by looking at the derivatives of the controls with regard to each parameter and is as expected on an intuitive level. We will therefore just summarize them. An increase in the difference between μ and r yields a better investment opportunity in either a long or short position in the risky asset. The same is true for a decrease in σ. A better investment opportunity will decrease consumption in favor of keeping the wealth in the market.

5.3. **Comparing with the Partial Problems.** We now return to the finite time setting. If we compare our problem to the investment-consumption problem without terminal value, this is the same as having the terminal condition $v(T,x) = 0$. Let the optimal consumption rate for this problem be denoted by $\tilde{c}$ and consider what would happen to (17) with the new terminal condition

$$\tilde{c}(t,x) = x\left(h(t) - exp\left\{\left(\frac{p(\mu-r)^2}{2\sigma^2(1-p)} + rp - \beta\right)(T-t)/(1-p)\right\}\right)^{-1}.$$

By definition the exponential function is strictly non-negative and we see that

$$\tilde{c} \geq \hat{c}\ ,\forall (t,x) \in [0,T] \times \mathbb{R}^+.$$

This is even more visible in the simulations below where we see a greater consumption and no terminal wealth.

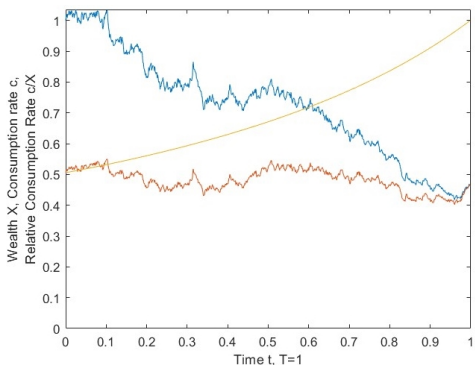

FIGURE 5.
$v(T,x) = exp\{-\beta t\}X_t^p$

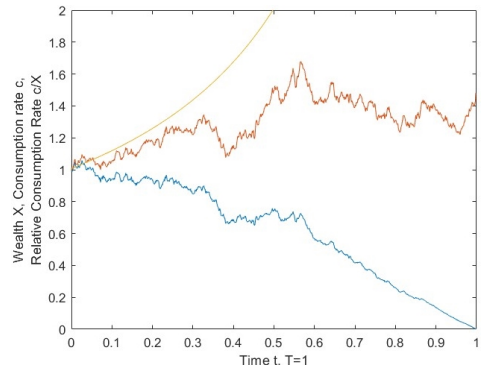

FIGURE 6.
$v(T,x) = 0$

Simulations of the problem with and without a bequest function
$p = 0.5,\ \beta = 0.03,\ \mu = 0.06,\ \sigma = 0.25,\ r = 0.02$

The difference in scale between the two figures is to get an idea of how big the relative consumption rate gets in Figure 6. The difference in consumption strategies is obvious on an intuitive level. If the investor does not care about the wealth at time T the consumption should be bigger up to time T, and everything should be consumed before time T.

We now look at the sub-problem without consumption, i.e. we want to know how the value of our composite portfolio Y develops over time compared to the case with consumption. As we see in the simulations below, we expect to have a greater terminal wealth at time T without consumption compared to with consumption. Which is in line with the intuition.

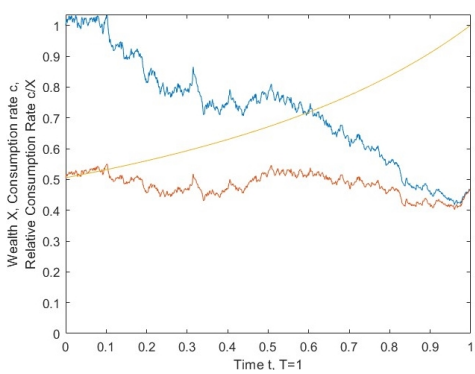

FIGURE 7.
With consumption

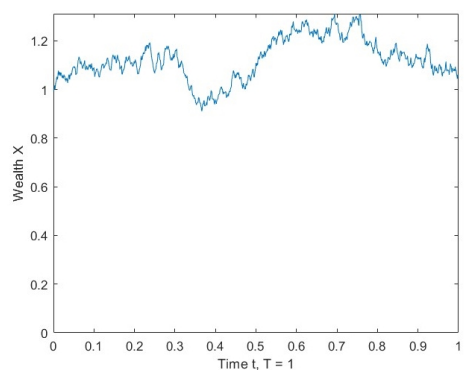

FIGURE 8.
Without consumption

Simulations of the problems with and without consumption
$p = 0.5,\ \beta = 0.03,\ \mu = 0.06,\ \sigma = 0.25,\ r = 0.02$

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References

[1] Richard Bellman. The theory of dynamic programming. *Bulletin of the American Mathematical Society*, 60(6):503–515, 1954.

[2] Tomas Björk. *Arbitrage Theory in Continuous Time*. Oxford Finance Series. Oxford University Press, fourth edition, 2019.

[3] Kiyosi Itô. 109. stochastic integral. *Proceedings of the Imperial Academy*, 20(8):519–524, 1944.

[4] The Wall Street Journal. Sweden 10 year government bond. `https://www.wsj.com/market-data/quotes/bond/BX/TMBMKSE-10Y`, 2022. [Online; accessed 09-June-2022].

[5] Chan-Chiao Lin, Huei Peng, and J.W. Grizzle. A stochastic control strategy for hybrid electric vehicles. In *Proceedings of the 2004 American Control Conference*, volume 5, pages 4710–4715 vol.5, 2004.

[6] Robert C Merton. Lifetime portfolio selection under uncertainty: The continuous-time case. *The review of Economics and Statistics*, pages 247–257, 1969.

[7] Robert C Merton. Optimum consumption and portfolio rules in a continuous-time model. In *Stochastic optimization models in finance*, pages 621–661. Elsevier, 1975.

[8] Donald J Meyer and Jack Meyer. Relative risk aversion: What do we know? *Journal of Risk and Uncertainty*, 31(3):243–262, 2005.

[9] NASDAQ Nordic. Historical prices omxs30, omx stockholm 30 index. `http://www.nasdaqomxnordic.com/indexes/historical_prices?Instrument=SE0000337842`, 2022. [Online; accessed 04-June-2022].

[10] Bernt Øksendal. *Stochastic Differential Equations: An Introduction with Applications*, pages 43–63. Springer Berlin Heidelberg, Berlin, Heidelberg, 2003.

[11] Adam E Parker. Who solved the bernoulli differential equation and how did they do it? *The College Mathematics Journal*, 44(2):89–97, 2013.

[12] Huyên Pham. *Continuous-time Stochastic Control and Optimization with Financial Applications*. Stochastic Modelling and Applied Probability. Springer Berlin Heidelberg, 2009.

[13] Walter Rudin. *Principles of mathematical analysis*. International Series in Pure and Applied Mathematics. McGraw-Hill Book Co., New York, third edition, 1976.

[14] Wenyu Zhang. Introduction to ito's lemma. *Lecture Note, Cornell University, Department of Statistical Sciences*, 2015.